

The Future of the Digital Camera



A few years ago, digital cameras were costly and rare gadgets. They had CMOS sensors that were almost as costly as a lens, and were capable of clicking images at not more 640 x 480 pixels. But in the last 2 to 3 years, the technology has advanced immensely. From a meagre 8 MB storage, some cameras now boast as much as 64 MB memory. Almost all of them support external memory cards, and can be plugged directly into a compatible photo printer that will print you photographs of excellent quality.

Earlier digital cameras lacked good image quality. This issue was resolved by replacing the CMOS with a CCD (charge coupled device). The CCD allows cameras to capture images at 10 megapixels. Cameras today are equipped with 10X zoom lens and can take even external SLR (Single Lens Reflex) lenses to make things better for professional users. In fact, newer digital cameras have all the features a professional would require.

The best part is that manufacturers can include as many feature as they wish because of the hardware and software combination. The normal 2 to 3 megapixel cameras have white balance settings, macro mode, red-eye reduction, video-clip recording, and 2X to 3X optical and up to 6X digital zoom. These features help even novices to capture brilliant images.

The wide acceptance of digital cameras and the ready availability of software to modify captured images have prompted newer operating systems such as Windows XP to bundle built-in driver support, making the devices virtually plug-and-play.

There are a few features found only in high-end cameras, which will surely be implemented in entry-level cameras soon. This includes TruePic technology from Olympus, which uses all the pixels even at lower resolutions. The software or algorithm has been written to enlarge the image even when it is in RAW format on the camera, to get a higher interpolated resolution than what's available with current cameras. This will reduce the artifacting that occurs while enlarging photos using image-editing software.

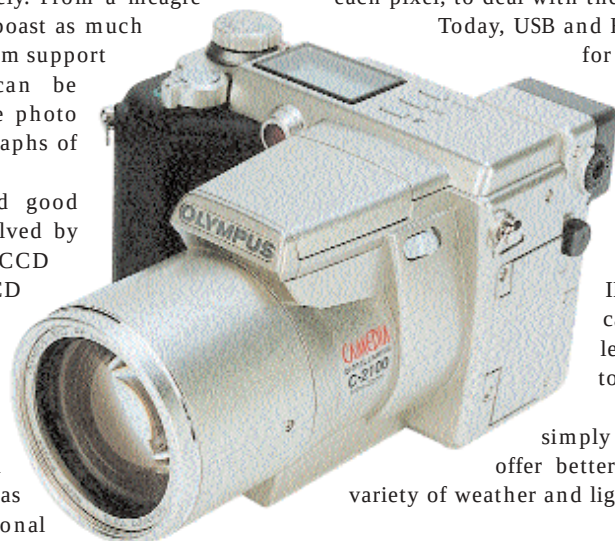
Future cameras will also have the lenses specifically designed to suit digital cameras. This is necessary because as the CCD count goes up, the density increases manifold. Even cameras equipped with the best CCD might not produce a great image if the lenses do not let the CCD see through it precisely when zooming in or out.

CCDs used in digital cameras have one very visible shortcoming as compared to film. Because of the way they function, CCDs cannot respond to the dynamic range—the range of light and dark values that can be recorded. Since CCDs are linear devices, the limit within which they can respond to the dynamic range is predefined: once they are fully charged they will not respond to any incremental exposure. Film is

more flexible in this regard and hence produces better results. But a new technology from Fuji promises to eliminate this problem. This is achieved by putting in one more low-sensitivity sensor to each pixel, to deal with the change in exposure levels.

Today, USB and FireWire are the interfaces of choice for most digital cameras. But as devices from laptops to PDAs, and even cell phones embrace wireless connectivity to 'talk' with the PC and other mobile devices, the digital camera will surely follow suit. Some gadgets such as the Casio wrist camera already have an IR port. And cell phones that have a camera also come with an IR port to let users download pictures wirelessly to their PC.

Digital cameras of the future won't simply be more feature-rich, they will also offer better photo quality and function in a variety of weather and lighting conditions.



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